

System Design Document

Design & Execution Workflow Management System

1. System Overview

The Design & Execution Workflow Management System is a simplified platform that supports all organizational roles while providing only essential workflow, task tracking, and collaboration features.

Core Purpose and Business Value

- Centralize project workflow across all departments.
- Improve communication between office and field teams.
- Provide basic visibility into project progress.
- Maintain structured collaboration across all roles.

Key Problems It Solves

- Fragmented communication across departments
- Lack of project visibility
- Poor coordination between design, finance, and site teams
- Unclear task ownership
- Difficulty tracking deadlines and deliverables

2. Users & Roles

All roles are included, but with **limited functionality**:

Marketing Lead

- Create new project leads
- Input client and project details

General Manager (GM)

- View all projects

- Provide feedback
- Approve/reject stages (single-level approval only)

Design Team Leader

- View assigned projects
- Assign tasks to designers

Designer

- View assigned tasks
- Upload design files

Site Engineer

- View project details
- Upload basic site data (optional/simple)

Finance Officer

- View project information
- Manually mark financial approval (no automation/workflow enforcement)

Purchasing Team

- View final project documents
- Access BOQ files (if uploaded)

System Administrator

- Manage users and roles
- Control system access

3. Features & Functionalities

Core Features (MVP Scope)

- Basic workflow management (manual stage updates)
- Task assignment & tracking
- Single-level approval system (GM only)
- File upload & download
- Deadline tracking (manual)

- Basic Role-Based Access Control (RBAC)

4. Core System Elements

Modules / Services

- **Authentication & Authorization Service**
Handles login and basic role validation
- **User Management Service**
Manages users, roles, and assignments
- **Project & Workflow Service**
Tracks project stages (manual transitions only)
- **Task & Assignment Service**
Assigns and tracks tasks
- **Approval Service (Basic)**
Supports only single-level approvals (GM)
- **Document Management Service**
Handles file upload/download (no version control)
- **Scheduling Service (Basic)**
Stores deadlines without automation

5. Security & Access Control

Authentication

- Username & password login

Authorization

- Basic role-based permissions

6. Limitations

- No workflow automation engine
- No document version control
- No real-time notifications

- No audit logs
- No performance analytics(Project velocity tracking)
- No Bottleneck detection
- No Designer performance metrics
- No Approval delays tracking
- No Management dashboards